AMC 10 · November Full Mock

Worked solutions • 25 questions • 75 minutes

Question 1 · Answer B

A print shop charges 8 credits to start a job and 3 credits per poster. Two groups pool orders of 7 and 9 posters into a single job. How many credits do they save?

Choices: A 6 B 8 C 3 D 16 E 24

The per-poster charges do not change. A pooled order uses one start charge instead of two, saving 8.

Answer: B · 8

Question 2 · Answer C

One rectangular flag is 3 units wider and 2 units shorter than a square flag of side s. Their areas are equal. What is s?

Choices: A 8 B 4 C 6 D 5 E 3

Expand $(s+3)(s-2)=s^2$ to obtain $s^2+s-6=s^2$. Therefore $s=6$.

Answer: C · 6

Question 3 · Answer E

A counter repeatedly cycles through 4, 1, 3, 2. What is the sum of its first 39 readings?

Choices: A 100 B 96 C 90 D 99 E 98

Nine full cycles contribute 90, and the next three terms contribute 4+1+3=8. Total 98.

Answer: E · 98

Question 4 · Answer C

A scale drawing shows a courtyard as a 3 cm by 5 cm rectangle. The actual perimeter is 64 m. What is its actual area in square metres?

Choices: A 120 B 256 C 240 D 192 E 60

The perimeter scale is 64/16=4 metres per centimetre. The actual sides are 12 and 20 m, giving area 240.

Answer: C · 240

Question 5 · Answer D

Five distinct positive integers have median 8 and maximum 14. What is the greatest possible sum of all five?

Choices: A 46 B 43 C 50 D 48 E 47

The two lower values can be at most 6 and 7; the value between 8 and 14 can be at most 13. Thus 6+7+8+13+14=48, which is attainable.

Answer: D · 48

Question 6 · Answer D

A four-digit number is made by writing the same two-digit number twice. What is the largest prime that divides every such four-digit number?

Choices: A 11 B 103 C 37 D 101 E 97

Repeating n produces 100n+n=101n. Consecutive n have no common prime factor, and 101 is prime. Hence 101 is the largest universal prime divisor.

Answer: D · 101

Question 7 · Answer A

Six different paintings go into two identical portfolios with three paintings in each. How many divisions are possible?

Choices: A 10 B 40 C 15 D 6 E 20

There are C(6,3)=20 ways to choose a named first portfolio. The portfolios are identical, so every division is counted twice: 10.

Answer: A · 10

Question 8 · Answer B

A moving walkway carries a standing passenger its full length in 90 seconds. A passenger walking at constant speed on the stationary walkway takes 60 seconds. How long does the walking passenger take when the walkway assists?

Choices: A 24 B 36 C 30 D 75 E 45

The combined rate is 1/90+1/60=1/36 walkway per second. One walkway takes 36 seconds.

Answer: B · 36

Question 9 · Answer C

Positive integers a and b satisfy ab+2a+2b=31. What is the largest possible value of a+b?

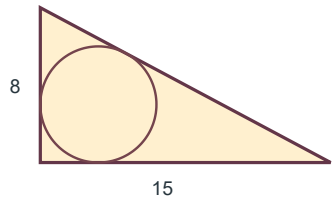
Choices: A 6 B 12 C 8 D 10 E 7

Add 4: (a+2)(b+2)=35. Both factors exceed 2, so they must be 5 and 7. Thus the unordered values are 3 and 5, of sum 8.

Answer: C · 8

Question 10 · Answer A

A circle is inscribed in a right triangle with legs 8 and 15. What is its radius?



Choices: A 3 B 2 C 15/4 D 4 E 5

The hypotenuse is 17. The area equals inradius times semiperimeter, so r=60/20=3.

Answer: A · 3

Question 11 · Answer C

Three fair coins are tossed. For each pair of coins showing the same face, one token is earned. What is the expected number of tokens?

Choices: A 9/4 B 2 C 3/2 D 1 E 3/4

There are three pairs. Each pair matches with probability 1/2, so linearity of expectation gives 3/2 tokens.

Answer: C · 3/2

Question 12 · Answer E

Six numbers in arithmetic progression have sum 51. The third and fifth differ by 6, with the fifth larger. What is the first number?

Choices: A 5 B 2 C 4 D 3 E 1

The common difference is 3. The mean of the six terms is 8.5 and equals the first term plus 2.5 differences. Thus the first term is 8.5−7.5=1.

Answer: E · 1

Question 13 · Answer A

A number N is the product 11 · 12 · … · 30. How many zeros are at the end of its decimal representation?

Choices: A 5 B 6 C 7 D 4 E 3

The number of 5 factors is one each from 15 and 20, two from 25, one from 30: five. There are more than five factors of 2, giving five trailing zeros.

Answer: A · 5

Question 14 · Answer D

A robot on a line makes five moves, each either +2 or −1. How many move strings end at 1, starting at 0, without ever going below 0?

Choices: A 5 B 2 C 10 D 3 E 4

To finish at 1 requires two +2 moves and three −1 moves. The first move must be +2. The second +2 can be in slot 2, 3 or 4; slot 5 would mean three drops first. Thus there are three strings.

Answer: D · 3

Question 15 · Answer C

The graph $y=|x-2|+|x+3|$ contains a horizontal segment. What is the product of its length and its height above the x-axis?

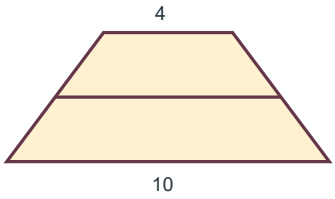
Choices: A 5 B 15 C 25 D 20 E 10

For $-3 \le x \le 2$ the distances to the endpoints sum to 5. Outside that interval the sum changes. The segment has length 5 and height 5, giving 25.

Answer: C · 25

Question 16 · Answer C

A trapezoid has parallel bases 4 and 10. The line joining the midpoints of its nonparallel sides cuts it into two pieces. What fraction of the total area belongs to the piece touching the shorter base?



Choices: A 4/11 B 17/28 C 11/28 D 1/2 E 2/7

The midpoint line has length 7; each piece has half the original height. The two areas are proportional to 4+7 and 7+10, so the required part is 11/28.

Answer: C · 11/28

Question 17 · Answer E

A school chooses representatives from eight clubs of six students each. Each student belongs to exactly one club. What is the fewest representatives that guarantee at least four come from the same club?

Choices: A 28 B 22 C 24 D 23 E 25

Without four from one club, at most three can come from each of eight clubs, for 24. That arrangement is possible. One more representative forces a fourth.

Answer: E · 25

Question 18 · Answer C

How many positive integers divide $2^6 \cdot 3^4 \cdot 5^2$ and have a perfect-square quotient when they divide that number?

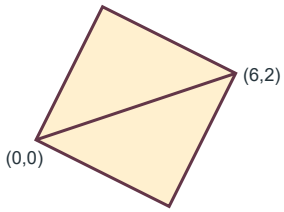
Choices: A 36 B 18 C 24 D 105 E 12

The quotient is square exactly when each remaining exponent is even. Exponents chosen in the divisor must therefore be even: 4 choices for 2, 3 for 3, and 2 for 5. Multiply to get 24.

Answer: C · 24

Question 19 · Answer C

The points (0,0) and (6,2) are opposite vertices of a square. What is the sum of the absolute values of the coordinates of both other vertices?



Choices: A 14 B 6 C 12 D 10 E 16

The midpoint is (3,1). Rotate the half-diagonal (3,1) through 90° to (−1,3). The other vertices are (2,4) and (4,−2). The four absolute values sum to 12.

Answer: C · 12

Question 20 · Answer B

Two fair six-sided dice are rolled. Given that at least one die shows an even number, what is the probability that their sum is divisible by 3?

Choices: A 4/9 B 1/3 C 1/4 D 2/9 E 1/2

Nine odd–odd pairs are excluded, leaving 27. Twelve of the original pairs have sum divisible by 3, of which three are odd–odd: (1,5),(3,3),(5,1). Thus 9/27=1/3.

Answer: B · 1/3

Question 21 · Answer E

Positive integers x,y satisfy 2x+y=29. What is the largest possible product xy?

Choices: A 108 B 98 C 100 D 104 E 105

The product is x(29−2x), a downward parabola with vertex at x=29/4. Compare neighbouring integers 7 and 8, giving 105 and 104. The maximum is 105.

Answer: E · 105

Question 22 · Answer C

Seven different musicians stand in a circle, with rotations identified and reflections distinct. Three specified musicians must stand consecutively as a single group. How many arrangements are possible?

Choices: A 120 B 720 C 144 D 240 E 72

Treat the trio as one block plus four single musicians: five objects on a circle give 4! arrangements. The trio can be ordered in 3! ways, for $24\times6=144$.

Answer: C · 144

Question 23 · Answer C

An integer sequence starts $a_1=2$ and $a_2=5$, with $a_{n+2}=2a_{n+1}-a_n+2$. Find a_{11}.

Choices: A 132 B 100 C 122 D 112 E 120

Successive differences start at 3 and increase by 2: 3,5,…,21. The ten differences sum to 120, so $a_{11}=122$.

Answer: C · 122

Question 24 · Answer C

How many ordered pairs of positive integers (x,y) satisfy $1/x+1/y=1/12$?

Choices: A 8 B 16 C 15 D 18 E 12

Rearrange to $(x-12)(y-12)=144$. Both factors are positive. Each positive divisor of $144=2^4\cdot3^2$ determines exactly one ordered pair; there are $5\times3=15$ divisors.

Answer: C · 15

Question 25 · Answer E

Eight points lie on a circle. Draw every chord joining two points. No three chords meet at a common interior point. Into how many regions do these chords divide the disk?

Choices: A 100 B 70 C 93 D 98 E 99

Each added chord creates one more region than the number of intersections it meets in its interior. There are $C(8,2)=28$ chords and $C(8,4)=70$ crossings, one per choice of four endpoints. Include the initial region: $1+28+70=99$.

Answer: E · 99
